SG No. 119 D_{2d}^{9} $I\bar{4}m2$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `2a`, site symmetry: `-4m2`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `-4m2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[1,2,-3,-4,5,6,-7,-8]`

Table 2: Wyckoff bond: `4b@2a`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[0,\ 0,\ 0]$	`[1,-2,5,-6]`
2	$[0,\ -X,\ 0]$	$[0,\ 0,\ 0]$	`[3,-4,-7,8]`

Table 3: Wyckoff bond: `4c@2a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X,\ X,\ 0]$	$[0,\ 0,\ 0]$	`[1,-2,7,-8]`
2	$[X,\ -X,\ 0]$	$[0,\ 0,\ 0]$	`[3,-4,5,-6]`

Table 4: Wyckoff bond: `8d@2a`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[1,5]`
2	$[-X,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[2,6]`
3	$[0,\ -X,\ -Z]$	$[0,\ 0,\ 0]$	`[3,8]`
4	$[0,\ X,\ -Z]$	$[0,\ 0,\ 0]$	`[4,7]`

Table 5: Wyckoff bond: `8e@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ Z]$	$[0,\ 0,\ 0]$	`[1,-8]`
2	$[-X,\ -X,\ Z]$	$[0,\ 0,\ 0]$	`[2,-7]`
3	$[X,\ -X,\ -Z]$	$[0,\ 0,\ 0]$	`[3,-6]`
4	$[-X,\ X,\ -Z]$	$[0,\ 0,\ 0]$	`[4,-5]`

Table 6: Wyckoff bond: `8f@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ 0]$	`[1,-2]`
2	$[Y,\ -X,\ 0]$	$[0,\ 0,\ 0]$	`[3,-4]`
3	$[X,\ -Y,\ 0]$	$[0,\ 0,\ 0]$	`[5,-6]`

4	[Y, X, 0]	[0, 0, 0]	`[7,-8]`

Table 7: Wyckoff bond: `16g@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	[X, Y, Z]	[0, 0, 0]	`[1]`
2	[$-X$, $-Y$, Z]	[0, 0, 0]	`[2]`
3	[Y, $-X$, $-Z$]	[0, 0, 0]	`[3]`
4	[$-Y$, X, $-Z$]	[0, 0, 0]	`[4]`
5	[X, $-Y$, Z]	[0, 0, 0]	`[5]`
6	[$-X$, Y, Z]	[0, 0, 0]	`[6]`
7	[Y, X, $-Z$]	[0, 0, 0]	`[7]`
8	[$-Y$, $-X$, $-Z$]	[0, 0, 0]	`[8]`

* Wyckoff site: `2b`, site symmetry: `-4m2`

Table 8: Wyckoff bond: `2a@2b`, bond symmetry: `-4m2`

No.	vector	center	mapping
1	[0, 0, Z]	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,-3,-4,5,6,-7,-8]`

Table 9: Wyckoff bond: `4b@2b`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	[X, 0, 0]	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,5,-6]`
2	[0, $-X$, 0]	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4,-7,8]`

Table 10: Wyckoff bond: `4c@2b`, bond symmetry: `2.22`

No.	vector	center	mapping
1	[X, X, 0]	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,7,-8]`
2	[X, $-X$, 0]	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4,5,-6]`

Table 11: Wyckoff bond: `8d@2b`, bond symmetry: `.m.`

No.	vector	center	mapping
1	[X, 0, Z]	$\left[0, 0, \frac{1}{2}\right]$	`[1,5]`
2	[$-X$, 0, Z]	$\left[0, 0, \frac{1}{2}\right]$	`[2,6]`
3	[0, $-X$, $-Z$]	$\left[0, 0, \frac{1}{2}\right]$	`[3,8]`
4	[0, X, $-Z$]	$\left[0, 0, \frac{1}{2}\right]$	`[4,7]`

Table 12: Wyckoff bond: `8e@2b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-7]`
3	$[X, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-6]`
4	$[-X, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4,-5]`

Table 13: Wyckoff bond: `8f@2b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2]`
2	$[Y, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4]`
3	$[X, -Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[7,-8]`

Table 14: Wyckoff bond: `16g@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2]`
3	$[Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3]`
4	$[-Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `2c`, site symmetry: `-4m2`

Table 15: Wyckoff bond: `2a@2c`, bond symmetry: `-4m2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,-3,-4,5,6,-7,-8]`

Table 16: Wyckoff bond: `4b@2c`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2,5,-6]`
2	$[0, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[3,-4,-7,8]`

Table 17: Wyckoff bond: `4c@2c`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2,7,-8]`

continued ...

Table 17

No.	vector	center	mapping
2	$[X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[3,-4,5,-6]`

Table 18: Wyckoff bond: `8d@2c`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,5]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,6]`
3	$[0, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[3,8]`
4	$[0, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[4,7]`

Table 19: Wyckoff bond: `8e@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,-7]`
3	$[X, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[3,-6]`
4	$[-X, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[4,-5]`

Table 20: Wyckoff bond: `8f@2c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2]`
2	$[Y, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[3,-4]`
3	$[X, -Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[7,-8]`

Table 21: Wyckoff bond: `16g@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2]`
3	$[Y, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[3]`
4	$[-Y, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[8]`

* Wyckoff site: `2d`, site symmetry: `-4m2`

Table 22: Wyckoff bond: `2a@2d`, bond symmetry: `-4m2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[1,2,-3,-4,5,6,-7,-8]`

Table 23: Wyckoff bond: `4b@2d`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[1,-2,5,-6]`
2	$[0, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[3,-4,-7,8]`

Table 24: Wyckoff bond: `4c@2d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[1,-2,7,-8]`
2	$[X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[3,-4,5,-6]`

Table 25: Wyckoff bond: `8d@2d`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[1,5]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[2,6]`
3	$[0, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[3,8]`
4	$[0, X, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[4,7]`

Table 26: Wyckoff bond: `8e@2d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[2,-7]`
3	$[X, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[3,-6]`
4	$[-X, X, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[4,-5]`

Table 27: Wyckoff bond: `8f@2d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[1,-2]`
2	$[Y, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[3,-4]`
3	$[X, -Y, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[7,-8]`

Table 28: Wyckoff bond: `16g@2d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[2]`
3	$[Y, -X, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[3]`
4	$[-Y, X, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[4]`
5	$[X, -Y, Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[5]`
6	$[-X, Y, Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[6]`
7	$[Y, X, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[8]`

* Wyckoff site: `4e`, site symmetry: `2mm.`

Table 29: Wyckoff bond: `4a@4e`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, z]$	`[1,-2,5,-6]`
2	$[0, -X, 0]$	$[0, 0, -z]$	`[3,-4,-7,8]`

Table 30: Wyckoff bond: `4b@4e`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2,5,6]`
2	$[0, 0, -Z]$	$[0, 0, -z]$	`[3,4,7,8]`

Table 31: Wyckoff bond: `8c@4e`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, z]$	`[1,5]`
2	$[-X, 0, Z]$	$[0, 0, z]$	`[2,6]`
3	$[0, -X, -Z]$	$[0, 0, -z]$	`[3,8]`
4	$[0, X, -Z]$	$[0, 0, -z]$	`[4,7]`

Table 32: Wyckoff bond: `8d@4e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[Y, -X, 0]$	$[0, 0, -z]$	`[3,-4]`
3	$[X, -Y, 0]$	$[0, 0, z]$	`[5,-6]`
4	$[Y, X, 0]$	$[0, 0, -z]$	`[7,-8]`

Table 33: Wyckoff bond: `16e@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[Y, -X, -Z]$	$[0, 0, -z]$	`[3]`
4	$[-Y, X, -Z]$	$[0, 0, -z]$	`[4]`
5	$[X, -Y, Z]$	$[0, 0, z]$	`[5]`
6	$[-X, Y, Z]$	$[0, 0, z]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, 0, -z]$	`[8]`

* Wyckoff site: `4f`, site symmetry: `2mm.`

Table 34: Wyckoff bond: `4a@4f`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2,5,-6]`
2	$[0, -X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,-4,-7,8]`

Table 35: Wyckoff bond: `4b@4f`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2,5,6]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,4,7,8]`

Table 36: Wyckoff bond: `8c@4f`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,5]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2,6]`
3	$[0, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,8]`
4	$[0, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[4,7]`

Table 37: Wyckoff bond: `8d@4f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,-4]`
3	$[X, -Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[7,-8]`

Table 38: Wyckoff bond: `16e@4f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2]`
3	$[Y, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3]`
4	$[-Y, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[8]`

* Wyckoff site: `8g`, site symmetry: `..2`

Table 39: Wyckoff bond: `8a@8g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[x, x, 0]$	`[1,-7]`
2	$[-X, X, Z]$	$[-x, -x, 0]$	`[2,-8]`
3	$[-X, -X, -Z]$	$[x, -x, 0]$	`[3,-5]`
4	$[X, X, -Z]$	$[-x, x, 0]$	`[4,-6]`

Table 40: Wyckoff bond: `8b@8g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$[x, x, 0]$	`[1,7]`
2	$[-X, -X, 0]$	$[-x, -x, 0]$	`[2,8]`
3	$[X, -X, 0]$	$[x, -x, 0]$	`[3,5]`
4	$[-X, X, 0]$	$[-x, x, 0]$	`[4,6]`

Table 41: Wyckoff bond: `16c@8g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, 0]$	`[2]`
3	$[Y, -X, -Z]$	$[x, -x, 0]$	`[3]`
4	$[-Y, X, -Z]$	$[-x, x, 0]$	`[4]`
5	$[X, -Y, Z]$	$[x, -x, 0]$	`[5]`
6	$[-X, Y, Z]$	$[-x, x, 0]$	`[6]`
7	$[Y, X, -Z]$	$[x, x, 0]$	`[7]`
8	$[-Y, -X, -Z]$	$[-x, -x, 0]$	`[8]`

* Wyckoff site: `8h`, site symmetry: `..2`

Table 42: Wyckoff bond: `8a@8h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	`[2,-8]`
3	$[-X, -X, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	`[3,-5]`
4	$[X, X, -Z]$	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	`[4,-6]`

Table 43: Wyckoff bond: `8b@8h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[1,7]`
2	$[-X, -X, 0]$	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	`[2,8]`
3	$[X, -X, 0]$	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	`[3,5]`
4	$[-X, X, 0]$	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	`[4,6]`

Table 44: Wyckoff bond: `16c@8h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	`[2]`
3	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	`[3]`
4	$[-Y, X, -Z]$	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	`[8]`

* Wyckoff site: `8i`, site symmetry: `.m.`

Table 45: Wyckoff bond: `8a@8i`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 0, z]$	`[1,5]`
2	$[-X, 0, Z]$	$[-x, 0, z]$	`[2,6]`
3	$[0, -X, -Z]$	$[0, -x, -z]$	`[3,8]`
4	$[0, X, -Z]$	$[0, x, -z]$	`[4,7]`

Table 46: Wyckoff bond: `8b@8i`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[0, X, 0]$	$[x, 0, z]$	`[1,-5]`
2	$[0, -X, 0]$	$[-x, 0, z]$	`[2,-6]`
3	$[X, 0, 0]$	$[0, -x, -z]$	`[3,-8]`
4	$[-X, 0, 0]$	$[0, x, -z]$	`[4,-7]`

Table 47: Wyckoff bond: `16c@8i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, z]$	`[2]`
3	$[Y, -X, -Z]$	$[0, -x, -z]$	`[3]`
4	$[-Y, X, -Z]$	$[0, x, -z]$	`[4]`
5	$[X, -Y, Z]$	$[x, 0, z]$	`[5]`
6	$[-X, Y, Z]$	$[-x, 0, z]$	`[6]`
7	$[Y, X, -Z]$	$[0, x, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, -x, -z]$	`[8]`

* Wyckoff site: `16j`, site symmetry: `1`

Table 48: Wyckoff bond: `16a@16j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[Y, -X, -Z]$	$[y, -x, -z]$	`[3]`
4	$[-Y, X, -Z]$	$[-y, x, -z]$	`[4]`
5	$[X, -Y, Z]$	$[x, -y, z]$	`[5]`
6	$[-X, Y, Z]$	$[-x, y, z]$	`[6]`
7	$[Y, X, -Z]$	$[y, x, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$[-y, -x, -z]$	`[8]`